■ System Requirements Specification (SRS) for Flutter App Project

■ 1. Development Environment Requirements

Hardware:

- Processor: Intel i5 (8th Gen or later) / AMD Ryzen 5 or higher
- RAM: Minimum 8 GB (16 GB recommended for smooth Android Studio/VS Code performance)
- Storage: At least 20 GB free (SSD recommended)
- GPU: For faster emulator performance (NVIDIA/AMD GPU recommended)

Software:

- OS: Windows 10/11 (64-bit) OR macOS (Monterey or later) OR Linux (Ubuntu 20.04 or later)
- Flutter SDK (latest stable release)
- Dart SDK (comes bundled with Flutter)
- Android Studio / VS Code with Flutter & Dart plugins
- Android SDK & Android Emulator
- Xcode (for iOS development, macOS only)
- Git (for version control)
- Firebase CLI (if using Firebase)
- Node.js (if integrating backend with certain tools)

■ 2. Application Functional Requirements

- User authentication (login/register with email, phone, or Google sign-in)
- Service browsing and search
- Booking & scheduling
- Push notifications (Firebase Cloud Messaging)
- Local storage (SQLite / Hive) for offline use
- Payment gateway integration
- Admin dashboard (web or mobile) for service providers

■ 3. Non-Functional Requirements

- Performance: App should load within 2–3 seconds, handle at least 1,000 concurrent users initially
- Security: Use HTTPS for all API calls, encrypted storage, JWT/Firebase Auth
- Usability: Intuitive UI, support dark mode, responsive design
- Compatibility: Android SDK 21+ (Lollipop), iOS 12.0+, target latest versions
- Scalability & Maintainability: DDD architecture + BLoC (or Riverpod), modular design

■ 4. Deployment Requirements

- Google Play Console account (for Android app publishing)
- Apple Developer Program account (for iOS app publishing)
- CI/CD pipeline (optional but recommended – e.g., GitHub Actions, Codemagic, Bitrise)